

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates $P(X \leq 2)$ or $P(X \leq 3)$ using the binomial dist	AO3.4	M1	$P(X \leq 2) = 0.55(177\dots)$ $P(X \leq 3) = 0.80(589\dots)$
	Obtains correct answer AWRT 0.448	AO1.1b	A1	$P(X \geq 3) = 1 - P(X \leq 2) = 0.448$
(b)	Calculates the cube of their answer to (a). Do not accept as part of a larger calculation or multiples of their cube.	AO1.1a	M1	$0.448\dots^3$ $= 0.090$ (3dp)
	Obtains correct answer (FT if AWWF 0.0899 to 0.0901)	AO1.1b	A1F	
(c)	States first appropriate assumption in context	AO3.5b	E1	The probability of hitting the bullseye is fixed at 0.3 or Hitting the bullseye with each dart is independent of hitting the bullseye with any other dart or There are 2 possible outcomes, hit bullseye or does not hit bullseye
	States second appropriate assumption in context Accept probability of hitting bullseye 'is constant' Do not accept 'fixed number of trials'	AO3.5b	E1	
Total 6 marks				

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States the correct binomial distribution	AO3.3	B1	$B(6, 0.15)$
(b)	Calculates the correct probability	AO1.1b	B1	0.0000114
(c)	Calculates $P(X \leq 1)$ or $P(X \leq 2)$ using the binomial distribution	AO1.1a	M1	$P(X \leq 1) = 0.7764$ $P(X \geq 2) = 1 - P(X \leq 1)$ $P(X \geq 2) = 0.224$
	Obtains the correct answer	AO1.1b	A1	

(d)	Finds the correct mean	AO1.1b	B1	0.9
(e)	States a first appropriate assumption in context	AO3.5b	B1	The probability of a light bulb being faulty is fixed
	States a second appropriate assumption in context	AO3.5b	B1	A light bulb being faulty is independent of any other light bulb being faulty
Total 7 marks				

Q3.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	0.26
Total 1 mark			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains probability from calculator	AO3.4	B1	$P(X \leq 2) = 0.678$
(i)	Obtains either of these figures (0.8791, 0.1074) PI	AO3.4	M1	$P(X \leq 3) = 0.8791$ $P(X = 0) = 0.1074$
(ii)	Obtains correct probability	AO1.1b	A1	$P(1 \leq X \leq 3) = 0.8791 - 0.1074$ $= 0.772$
(b)	Recalls correct name for sampling method	AO1.2	B1	Opportunity sampling
(i)	States that sampling method is unrepresentative giving one appropriate weakness	AO3.5b	E1	The 25 students all come from the same college and cannot be said to fairly represent all students. There could be a regional difference in diet.
(ii)	States both hypotheses using correct notation	AO2.5	B1	$H_0: p = 0.2$ $H_1: p > 0.2$
(iii)	States or uses B(25, 0.2) PI	AO3.3	M1	Under H_0 , use $X \sim B(25, 0.2)$ (where X represents number of students eating

			5 or more portions)
Obtains correct probability	AO1.1b	A1	$P(X \geq 8) = 0.109$
Evaluates model by comparing $P(X \geq 8)$ with 0.05 (condone 0.0468/0.047 used instead of 0.109)	AO3.5a	M1	$0.109 > 0.05$
Infers H_0 accepted	AO2.2b	A1	Hence accept H_0
States correct conclusion in given context	AO3.2a	E1	No significant evidence that more than 20% eat at least five a day
Total 11 marks			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds value for p	AO1.1a	M1	$p = \frac{1680}{2400} = 0.7$
	Finds correct probability from calculator	AO1.1b	A1	Using $X \sim B(25, 0.7)$, $P(X = 22) = 0.0243$
(b)	Explains the reason why the model may no longer apply in context	AO3.5b	E1	It is likely that all the houses (and gardens) will be of similar types, and hence similar owners, so not likely to be independent as binomial model requires.
Total 3 marks				

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds correct value of k	AO1.1b	B1	$k = \frac{1}{16}$
(b)	Selects relevant probability	AO1.1a	M1	$P(\geq 2 \text{ checkouts staffed})$
	Finds correct probability FT 'their' value of k found	AO1.1b	A1F	$\frac{3}{16} + k = \frac{3}{16} + \frac{1}{16} = \frac{1}{4}$

in part (a)			ALT $P(\geq 2 \text{ checkouts staffed})$ $= 1 - \frac{3}{4} = \frac{1}{4}$
Total 3 marks			

Q7.

(a) $E \sim B(40, 0.30)$

*Used anywhere in question even by implication from a **correct** value*

M1

$P(E \leq 10) = \underline{\mathbf{0.308 \text{ to } 0.309}}$

AWFW (0.3087)

A1

(2)

(b) $P(E \geq 15) = \underline{\mathbf{1 - (0.8074 \text{ or } 0.8849)}}$

*Requires '1 - '
 Accept 3 dp rounding or truncation
 Can be implied by 0.192 to 0.193
 but **not** by 0.115 to 0.116*

M1

$= \underline{\mathbf{0.192 \text{ to } 0.193}}$

AWFW (0.1926)

A1

(2)

(c) $P(6 \leq E \leq 18) = (\underline{\mathbf{0.9852 \text{ or } 0.9680}})$

Accept 3 dp rounding or truncation

M1

MINUS $(\underline{\mathbf{0.0086 \text{ or } 0.0238}})$

Accept 3 dp rounding or truncation

M1

$= \underline{\mathbf{0.976 \text{ to } 0.977}}$

AWFW (0.9766)

A1

(3)

[7]

Q8.

- (a) (i) Volume. $X \sim N(507.5, 4.0^2)$

$$P(X < 515) = P\left(Z < \frac{515 - 507.5}{4.0}\right)$$

Standardising 515 with 507.5 and 4.0 but allow (507.5 - 515)

M1

$$P(Z < 1.875) = \underline{\mathbf{0.97}}$$

AWRT

(0.96960)

A1

- (ii) $P(500 < X < 515) = P(-a < Z < a) =$

$$P(Z < a) - (1 - P(Z < a))$$

or

$$2 \times P(Z < a) - 1$$

*OE; $a = 1.875$, $p = 0.97$ (AWRT) or correct standardising are not required May be implied by a **correct** answer*

M1

$$= [(i) - (1 - (i)) \text{ or } (2 \times (i)) - 1] = \underline{\mathbf{0.94}}$$

AWRT

(0.93921)

A1

- (iii) $P(X \neq 507.5) = \underline{\mathbf{1 \text{ or one or unity or } 100\%}}$

CAO; accept nothing else but ignore additional words providing that they are not contradictory (eg certain so = 1)

B1

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5

- (b) $0.96 \Rightarrow z = \underline{\mathbf{1.75}}$

AWRT (ignore sign)(1.7507)

B1

$$\left(\frac{x - 507.5}{4.0}\right) = 1.5 \text{ to } 2.1$$

*Standardising x with 507.5 and 4.0 but allow (507.5 - x); **and** equating to a **z-value** (ignore signs)*

M1

$$x = \underline{\mathbf{514 \text{ to } 515}}$$

AWFW

(514.50)

Must be consistent signs throughout

A1

3

(c) $p = \underline{(a)(ii)} [= 0.94]$

F on (a)(ii) providing $0 < p < 1$

BF1

$$P(\geq 5 \text{ in } 6) = (6 \times p^5 \times (1 - p)) + p^6$$

Either expression with value of p

M1

$$= \underline{0.948 \text{ to } 0.956}$$

AWFW

(0.95296)

A1

3

[11]

Q9.

(a) (i) $O \sim B(40, p)$

Accept percentage equivalents except for 27

$$P(NS \leq 10) = \underline{0.97}$$

AWRT

(0.9701)

B1

1

(ii) $P(LPE \geq 25) = \underline{1 - (0.9231 \text{ or } 0.9597)}$

Requires '1 - '

Accept 3 dp rounding

*Can be implied by (0.0769 to 0.077) but **not** by (0.04 to 0.0403)*

M1

$$= \underline{0.077}$$

AWRT

(0.0769)

A1

2

(iii) $P(UPE = 2) = \binom{40}{2} (0.175)^2 (0.825)^{38}$

*Correct expression; may be implied by a **correct** answer*

Ignore extra terms

M1

$$= \underline{\mathbf{0.016}}$$

AWRT (0.0160)

A1

2

(iv) $p = 0.85 - 0.50 = \underline{\mathbf{0.35}}$

CAO; award on value only May be implied by any of four probabilities below or by a **correct** answer

B1

$$P(10 < X < 15) = \mathbf{0.5721} \text{ or } \mathbf{0.6946} \text{ } (p_1)$$

Accept 3 dp rounding

May be implied by a **correct** answer

M1

$$\mathbf{MINUS} \quad \mathbf{0.1215} \text{ or } \mathbf{0.0644} \text{ } (p_2)$$

Accept 3 dp rounding

May be implied by a **correct** answer

M1

$$= \underline{\mathbf{0.45 \text{ to } 0.451}}$$

AWFW (0.4506)

A1

4

(b) $p = 0.85 - 0.175 = \underline{\mathbf{0.675}}$

CAO; may be implied by 27

Each can be found in several ways

or

$$p' = \underline{\mathbf{0.325}}$$

CAO; may be implied by 13 or 27

B1

$$\text{Number} = 40 \times 0.675 = \underline{\mathbf{27}}$$

CAO; can be found in several ways

B1

2

[11]

Q10.

(a) $U \sim B(40, 0.15)$

Used somewhere in (a)

M1

(i) $P(U = 6) = 0.6067 - 0.4325$
or

$$= \binom{40}{6} (0.15)^6 (0.85)^{34}$$

Accept 3 dp rounding or truncation

M1

$$= 0.174$$

Can be implied by a correct answer

AWRT (0.1742)

A1

3

(ii) $P(U \leq 5) = 0.432 \text{ to } 0.433$

AWFW (0.4325)

B1

1

(iii) $P(5 < U < 10) = 0.9328 \text{ or } 0.9701$ (p1)

Accept 3 dp rounding or truncation but allow 0.97

M1

$$p_2 - p_1 \Rightarrow M0 M0 A0$$

$$(1 - p_2) - p_1 \Rightarrow M0 M0 A0$$

$$p_1 - (1 - p_2) \Rightarrow M0 M0 A0$$

$$(1 - p_2) - (1 - p_1) \Rightarrow M1 M1 (A1)$$

only providing result > 0

MINUS 0.4325 or 0.2633 (p2)

Accept 3 dp rounding or truncation

M1

$$= 0.5(00) \text{ to } 0.501$$

AWFW (0.5003)

A1

3

Alternative solution

B(40, 0.15) expressions stated for at least 3 terms within
 $5 \leq U \leq 10$ gives probability

Can be implied by a correct answer

(M2)

= 0.5(00) to 0.501

AWFW

(0.5003)

(A1)

u	(5)	6	7	8	9	(10)
$P(U = u)$	(0.1692)	0.1742	0.1492	0.1087	0.0682	(0.0373)

(3)

(b) Mean or $\mu = 32 \times 0.15 = 4.8$

CAO

B1

(V or σ^2) = $\frac{32 \times 0.15 \times 0.85}{}$

or

(SD or σ) = $\frac{\sqrt{32 \times 0.15 \times 0.85}}{}$

Either numerical expression; ignore terminology

May be implied by 4.08 CAO seen or 2.02 AWRT seen

M1

(SD or σ) = 2.02

AWRT

(2.0199)

Do not award if labelled V or σ^2

A1

3

(c) Mean = 7.7

CAO

($\sum x - 77$)

B1

SD = 1.26 to 1.34

AWFW

($\sum x^2 = 609$)

B1

(Sample) mean is bigger / greater / different or $7.7 / 32 = 0.24 > 0.15$
 and

(Sample) SD is smaller / less / different

Both; dependent on all previous 5 marks of B1 M1 A1 B1 B1
Can be scored for incorrect (b) re-done correctly in (c)
Means & SDs different → Bdep0

Bdep1

So model appears unsuitable
 OE; dependent on Bdep1

Bdep1

4

[14]

Q11.

(a) $B(n, p)$

Used correctly anywhere in question

M1

$$P(C = 2) = \binom{10}{2} (0.275)^2 (0.725)^8$$

*Can be implied by a **correct** answer*

A1

$$= \underline{\underline{0.259 \text{ to } 0.26}}$$

AWFW (0.25977)

A1

3

(b) $n = \underline{\underline{40}}$ and $p = \underline{\underline{0.35}}$

CAO both; used in (b)

B1

(i) $P(G \leq 15) = \underline{\underline{0.694 \text{ to } 0.695}}$

AWFW (0.6946)

B1

2

(ii) $P(10 \leq G \leq 20) = \underline{\underline{0.9827 \text{ or } 0.9637}}$ (p_1)

Accept 3 dp rounding or truncation

M1

MINUS 0.0644 or 0.1215 (p_2)

Accept 3 dp rounding or truncation

M1

$$= \underline{0.918 \text{ to } 0.92} (p_3)$$

AWFW (0.9183)

Notes 1 $p_3 \leq 0$ or $p_3 \geq 1 \Rightarrow$ M0 M0 A0

2 $p_2 - p_1 \Rightarrow$ M0 M0 A0

3 $(1 - p_2) - p_1 \Rightarrow$ M0 M0 A0

4 $p_1 - (1 - p_2) \Rightarrow$ M1 M0 A0

5 $p_1 \times p_2 \Rightarrow$ M1 M0 A0

6 $(1 - p_2) - (1 - p_1) \Rightarrow$ M1 M1 (A1)

A1

3

(c) $n = \underline{50}$ and $p = \underline{0.85}$

$$P(R > 40 | p = 0.85) = P(A < 10 | p = 0.15)$$

Attempt to change from R to A

M1

$$= \underline{0.7911 \text{ or } 0.8801}$$

Accept 3 dp rounding or truncation

m1

$$= \underline{0.79 \text{ to } 0.792}$$

AWFW

(0.7911)

A1

OR

$$P(R \geq 40 | p = 0.85) = \underline{0.2089 \text{ or } 0.1199}$$

Accept 3 dp rounding or truncation

(M1)

$$P(R > 40 | p = 0.85) = \underline{1 - (0.2089 \text{ or } 0.1199)}$$

Requires '1 -'

Can be implied by 0.79 to 0.792

*but **not** by 0.88 to 0.881*

(m1)

$$= \underline{0.79 \text{ to } 0.792}$$

AWFW

(0.7911)

(A1)

3

[11]

Q12.

In (a)(i) & (c), ignore the inclusion of a lower limit of 0; it has no effect on either answer

(a) Weight, $W \sim N(2.75, 0.15^2)$

$$(i) \quad P(W < 2.8) = P\left(Z < \frac{2.8 - 2.75}{0.15}\right)$$

Standardising 2.8 with 2.75 and 0.15; allow (2.75 - 2.8)

M1

$$= P(Z < \underline{0.33 \text{ or } 1/3})$$

*AWRT / CAO; ignore inequality and sign
May be implied by a **correct** answer*

A1

$$= \underline{0.629 \text{ to } 0.633}$$

AWFW (0.63056)

A1

$$(ii) \quad P(W > 2.5) = P(Z > -1.67) = P(Z < +1.67)$$

Correct area change

*May be implied by a **correct** answer
or an answer > 0.5*

M1

$$= \underline{0.951 \text{ to } 0.953}$$

AWFW (0.95221)

A1

5

(b) Weight, $X \sim N(5.25, 0.20^2)$

$$(i) \quad P(5.1 < X < 5.3) = P(Z < 0.25) - P(Z < -0.75)$$

Must have diff of 2 probs for each B1

$$= \underline{0.59871}$$

Accept 0.599

B1

MINUS [(1 - 0.77337) or 0.22663] = 0.372(08)

Accept 0.773 or 0.227

*AG; do **not** mark simply on answer*

B1

2

(ii) $P(0 \text{ in } 4) = [1 - 0.372]^4$

Accept [1 - c's (b)(i)]⁴

M1

= 0.628⁴ = 0.155 to 0.156

AWFW (0.15554)

A1

2

[9]

Q13.

(a) (i) $U \sim B(30, 0.13, 0.35 \text{ or } 0.20)$

Used correctly anywhere in (a)

M1

$P(P = 2) = \binom{30}{2} (0.13)^2 (0.87)^{28}$

*Can be implied by a **correct** answer*

A1

= 0.148 to 0.15

AWFW (0.1489)

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A1

3

(ii) $p = \underline{\mathbf{0.35}}$

CAO

B1

$P(R \cup P > 10) = \underline{\mathbf{1 - (0.5078 \text{ or } 0.3575)}}$

Requires '1 -'

Accept 3 dp rounding or truncation

Can be implied by 0.49 to 0.493

*but **not** by 0.642 to 0.643*

M1

= 0.49 to 0.493

AWFW (0.4922)

A1

3

(iii) $P(5 \leq G \leq 10) = \mathbf{0.9744 \text{ or } 0.9389}(p_1)$

Accept 3 dp rounding or truncation

M1

MINUS 0.2552 or 0.4275 (p_2)

Accept 3 dp rounding or truncation

M1

= 0.719 to 0.72 (p_3)

AWFW (0.7192)

A1

3

Notes 1 $p_3 \leq 0$ or $p_3 \geq 1 \Rightarrow$ M0 M0 A0

2 $p_2 - p_1 \Rightarrow$ M0 M0 A0

3 $(1 - p_2) - p_1 \Rightarrow$ M0 M0 A0

4 $p_1 - (1 - p_2) \Rightarrow$ M1 M0 A0

5 $p_1 \times p_2 \Rightarrow$ M1 M0 A0

6 $(1 - p_2) - (1 - p_1) \Rightarrow$ M1 M1 (A1)

- (b) (i) Mean or $\mu = 100 \times 0.22 = \mathbf{22}$
 Variance or $\sigma^2 = 100 \times 0.22 \times 0.78$
 CAO

B1

= 17.1 to 17.2

AWFW (ignore notation) (17.16)

ISW all subsequent working

B1

2

- (ii) 22.1 $\approx$ / = 22 or means similar / equal **or**
 0.221 $\approx$ / = 0.22 or proportions similar / equal so
reject claim (that $p > 0.22$)

or

accept that $p = 0.22$

Dependent on 22 seen in (b)(i) or (ii)

Accept diff = 0.1 CAO

Correct (numerical)

comparison with **correct** conclusion (even if at end and stated as 'reject (both) claims')

B1

$$\sqrt{17.1 \text{ to } 17.2} = \underline{4.13 \text{ to } 4.15} \approx / = \underline{4.17}$$

Comparison using two values **or**
 one value + diff (0.02 to 0.04 AFWW)

or

$$\underline{17.1 \text{ to } 17.2} \approx / = \underline{17.3 \text{ to } 17.4}$$

Comparison using two values **or**
 one value + diff (0.1 to 0.3 AFWW)

B1

so

reject claim that not random samples

or

accept that are random samples

Dependent on previous B1

Correct conclusion regarding
 randomness of sample

Bdep1

3

[14]

Q14.

(a) $R \sim (15, 0.45)$

(i) $P(R \leq 5) = 0.26$ (0) to 0.261
 AFWW (0.2608)

B1

1

(ii) $P(R > 10) = 1 - P(R \leq 10)$

Requires '1 -'

Accept 3dp rounding or truncation

$$= 1 - (0.9745 \text{ or } 0.9231)$$

*Can be implied by 0.025 to 0.026 but not
 by 0.0769 to 0.077*

M1

$$= 0.025 \text{ to } 0.026$$

AWFW (0.0255)

A1

2

(iii) $P(R = 6) = 0.4522 - (a)(i) \text{ or } = \binom{15}{6} (0.45)^6 (0.55)^9$

Can be implied by a correct answer

M1

$$= 0.191 \text{ to } 0.192$$

AWFW (0.1914)

A1

2

- (iv) $P(5 \leq R \leq 10) = 0.9745 \text{ or } 0.9231 (p_1)$
Accept 3dp rounding or truncation
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

Minus $0.1204 \text{ or } 0.2608(p_2)$
Accept 3dp rounding or truncation

M1

$$= 0.853 \text{ to } 0.855$$

AWFW (0.8541)

A1

3

Or

B (15, 0.45) terms stated for at least 3 values within
 $4 R \leq 11$ gives probability

Can be implied by a correct answer

(M1)

$$= 0.853 \text{ to } 0.855$$

AWFW (0.8541)

(A2)

- (b) (i) $P(S) = 0.85 \text{ plus } 1 \text{ minus}$
CAO; requires 'plus' or 'minus'

B1

$$(0.15 \times 0.80) \quad (0.15 \times 0.20)$$

CAO; not simply 0.12 or 0.03

B1

$$= 0.97$$

AG

2

NB: $(0.85 \times 0.20) + 0.80 \Rightarrow$ B0 B0
 $(0.85 \times 0.20) + (0.85 \times 0.80) + (0.15 \times 0.80) \Rightarrow$ B0 B1

- (ii) $P(S \geq 48) = 0.81$ to 0.82 or 0.5553 or 0.9372
Accept 3dp rounding or truncation
M2 for the three correctly expressed terms
for B (50, 0.03) or B (50, 0.97) added

M2

$= 0.81(0)$ to 0.811
AWFW (0.8108)

A1

3

NB: Answer = 0.4447 or 0.1892 or $0.0628 \Rightarrow$ M1 only

- (iii) $p = 1 - 0.85 = 0.15$
CAO; may be implied by correct answer
or correct expression for mean

B1

Mean, $\mu = 80 \times 0.15 = 12$
 CAO

B1

2

SC Mean = $9.6 \Rightarrow$ B1 only

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[15]

Q15.

(a) $X \sim B(10, 0.15)$

(i) $P(X \leq 2) = \mathbf{0.82(0)}$
AWRT *(0.8202)*

B1

1

(ii) $P(X \geq 2) = 1 - P(X \leq 1)$

$= \mathbf{1 - (0.5443 \text{ or } 0.8202)}$

Requires '1 -'

Accept 3/2 dp rounding or truncation Can be implied by
*0.455 to 0.456 but **not** by 0.179 to 0.18(0)*

M1

$$= 0.455 \text{ to } 0.456$$

AWFW (0.4557)

A1

2

(iii) $P(1 < X < 5) = 0.9901 \text{ or } 0.9986 \quad (p_1)$
 Accept 3 dp rounding or truncation

$$p_2 - p_1 \Rightarrow M0 \ M0 \ A0$$

$$(1 - p_2) - p_1 \Rightarrow M0 \ M0 \ A0$$

$$p_1 - (1 - p_2) \Rightarrow M1 \ M0 \ A0$$

only providing result > 0

M1

minus 0.5443 or 0.1969 (p_2)
 Accept 3 dp rounding or truncation

M1

$$= 0.445 \text{ to } 0.446$$

AWFW (0.8541)

A1

OR

B(10, 0.15) expressions stated for **at least 3** terms within
 $1 \leq X \leq 5$ gives probability
 Can be implied by a correct answer

(M1)

$$= 0.445 \text{ to } 0.446$$

AWFW (0.8541)

(A2)

3

(b) $Y \sim B(50, 0.15)$

(i) $P(Y > 5) = 1 - P(Y \leq 5)$

$$= 1 - (0.2194 \text{ or } 0.1121)$$

Requires '1 -'

Accept 3 dp rounding or truncation Can be implied by
 0.78(0) to 0.781 but **not** by 0.888 to 0.89

M1

$$= 0.78(0) \text{ to } 0.781$$

AWFW (0.7806)

A1

2

(ii) $P(5 \leq Y \leq 10) = \mathbf{0.8801 \text{ or } 0.7911} \quad (p_1)$

Accept 2/3 dp rounding or truncation

$$p_2 - p_1 \Rightarrow M0 \ M0 \ A0$$

$$(1 - p_2) - p_1 \Rightarrow M0 \ M0 \ A0$$

$$p_1 - (1 - p_2) \Rightarrow M1 \ M0 \ A0$$

only providing result > 0

M1

minus 0.1121 or 0.2194 (p_2)

Accept 3 dp rounding or truncation

M1

= 0.768

AWRT

(0.7680)

A1

OR

B(50, 0.15) expressions stated for **at least 3** terms within
 $4 \leq Y \leq 10$ gives probability

Can be implied by a correct answer

(M1)

= 0.768

AWRT

(0.7680)

(A2)

3

(c) Mean, $\mu = 50 \times 10 \times 0.15 = \mathbf{75}$

CAO

SC: 7.5 & 6.37 to 6.38 $\Rightarrow B1$

B1

Variance, $\sigma^2 = 50 \times 10 \times 0.15 \times 0.85$

= 63.7 to 63.8

AWFW

(63.75)

B1

2

[13]

Q16.

(a) (i) Diameter, $D \sim N(57.15, 0.04^2)$

$$P(D < 57.2) = P\left(Z < \frac{57.2 - 57.15}{0.04}\right)$$

Standardising 57.2 with 57.15 and 0.04; allow (57.15 – 57.2)

M1

$$= P(Z < \mathbf{1.25})$$

*CAO; ignore inequality and sign
May be implied by a **correct** answer*

A1

$$= \mathbf{0.894 \text{ to } 0.895}$$

AWFW (0.89435)

A1

3

(ii) $P(57.1 < D < 57.2)$

$$= p - (1 - p)$$

*Allow even if incorrect standardising providing
 $p - (1 - p)$ **seen**
May be implied by a correct answer*

M1

$$= 2 \times 0.89435 - 1 = \mathbf{0.788 \text{ to } 0.79(0)}$$

AWFW (0.78870)

A1

2

(b) $P(16 \text{ balls} < 57.2) = p^{16}$ with $0 < p < 1$

*Any probability to power 16 or $1 - p^{16}$;
do **not** allow multiplying factors
If only seen in (b)(ii), allow just M1*

M1

$$= [(a)(i)]^{16} = (0.89435)^{16} = \mathbf{0.166 \text{ to } 0.17(0)}$$

AWFW (0.16754)

A1

2

[7]

Q17.

(a) $X \sim B(10, 0.15)$

(i) $P(X \leq 2) = \mathbf{0.82(0)}$

AWRT (0.8202)

B1

1

(ii) $P(X \geq 2) = 1 - P(X \leq 1)$

$= 1 - (0.5443 \text{ or } 0.8202)$

Requires '1 -'

Accept 3/2 dp rounding or truncation

*Can be implied by 0.455 to 0.456 but **not** by 0.179 to 0.18(0)*

M1

$= 0.455 \text{ to } 0.456$

AWFW (0.4557)

A1

2

(iii) $P(1 < X < 5) = 0.9901 \text{ or } 0.9986 \quad (p_1)$

Accept 3 dp rounding or truncation

$p_2 - p_1 \Rightarrow M0 M0 A0$

$(1 - p_2) - p_1 \Rightarrow M0 M0 A0$

$p_1 - (1 - p_2) \Rightarrow M1 M0 A0$

only providing result > 0

M1

minus $0.5443 \text{ or } 0.1969 \quad (p_2)$

Accept 3 dp rounding or truncation

M1

$= 0.445 \text{ to } 0.446$

AWFW (0.4458)

A1

OR

B(10, 0.15) expressions stated for **at least 3** terms within
 $1 \leq X \leq 5$ gives probability

Can be implied by a correct answer

(M1)

$= 0.445 \text{ to } 0.446$

AWFW (0.4458)

(A2)

3

(b) $Y \sim B(50, 0.15)$

Normal approximation $\Rightarrow$ 0 marks

(i) $P(Y > 5) = 1 - P(Y \leq 5)$

$= 1 - (0.2194 \text{ or } 0.1121)$

Requires '1 -'

*Accept 3 dp rounding or truncation Can be implied by
0.78(0) to 0.781 but **not** by 0.888 to 0.89*

M1

$= 0.78(0) \text{ to } 0.781$

AWFW

(0.7806)

A1

2

(ii) $P(5 \leq Y \leq 10) = 0.8801 \text{ or } 0.7911 \quad (p_1)$

Accept 2/3 dp rounding or truncation

$p_2 - p_1 \Rightarrow M0 M0 A0$

$(1 - p_2) - p_1 \Rightarrow M0 M0 A0$

$p_1 - (1 - p_2) \Rightarrow M1 M0 A0$

only providing result > 0

M1

minus 0.1121 or 0.2194 (p_2)

Accept 3 dp rounding or truncation

M1

$= 0.768$

AWRT

(0.7680)

A1

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OR

B(50, 0.15) expressions stated for **at least 3** terms within
 $4 \leq Y \leq 10$ gives probability

Can be implied by a correct answer

(M1)

$= 0.768$

AWRT

(0.7680)

(A2)

3

[11]

Q18.

- (a) (i) $R \sim B(14, 0.35)$
Used in (a); may be implied

M1

$$P(R \leq 7) = 0.924 \text{ to } 0.925$$

AWFW (0.92466)

A1

2

- (ii) $P(R \geq 11) = 1 - P(R \leq 10)$
 $= 1 - (0.9989 \text{ or } 0.9999)$
Requires '1 -' and ≥ 4 dp accuracy

M1

$$= 0.0011$$

AWRT (0.001106)

A1

2

- (iii) $P(5 < R < 10) = 0.9940 \text{ or } 0.9989 (p_1)$
Accept 3 dp accuracy
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

minus 0.6405 or 0.4227 (p_2)
Accept 3 dp accuracy

M1

$$= 0.353 \text{ to } 0.354$$

AWFW (0.35346)

A1

or
 $B(14, 0.35)$ expressions stated for **at**
least 3 terms within $4 \leq R \leq 11$ gives probability
Can be implied by correct answer

(M1)

$$= 0.353 \text{ to } 0.354$$

AWFW (0.35346)

(A2)

3

(b) $R \sim B(21, 0.35)$

Implied from correct stated formula; do not accept misreads

M1

$$P(R = 4) = \binom{21}{4} (0.35)^4 (0.65)^{17}$$

*Can be implied by a correct answer
Ignore any additional terms*

A1

$$= 0.059 \text{ to } 0.0595$$

AWFW (0.059274)

A1

3

[10]

Q19.

(a) (i) $X \sim N(10.2, 0.15^2)$

$$P(X < 10.5) = P\left(Z < \frac{10.5 - 10.2}{0.15}\right)$$

Standardising (10.45, 10.5 or 10.55) with 10.2

and ($\sqrt{0.15}$, 0.15 or 0.15^2) and/or $(10.2 - x)$

M1

$$= P(Z < 2)$$

*CAO; ignore inequality and sign
May be implied by a correct answer*

A1

$$= 0.977$$

AWRT (0.97725)

A1

3

(ii) $P(10.0 < X < 10.5)$

$$= [C's (a)(i)] - P(X < 10.0)$$

*Or equivalent; must be clear correct
method if answer incorrect and*

answer > 0

M1

$$= (a)(i) - P(Z < -1.33)$$

$$= (a)(i) - (1 - p)$$

Method correct using - 1.3

gives 0.88 to 0.881

$\Rightarrow$ M1 m1 A0

$$= 0.97725 - (1 - 0.90824)$$

Area change

May be implied by a correct answer or

answer > 0.5

m1

$$= 0.885 \text{ to } 0.887$$

AWFW (0.88604)

A1

M1 m1 A1 for $0.90824 - [1 - (a)(i)] = 0.886$

M1 m0 A0 for $(a)(i) - 0.90824 = 0.0685$

M0 m0 A0 for answer < 0

3

(b) $P(X > 10) = p[\text{from (a)(ii)}]$

$$= 0.908 \text{ to } 0.909$$

Correct value or F on value used or implied in (a)(ii) providing > 0.5

Use of - 1.3 gives 0.9032

B1F

$$P(6 \text{ rolls} > 10) = 0.90824^6$$

Accept any probability to power 6

M1

$$0.56 \text{ to } 0.565$$

AWFW

A1

Note:

B0F M1 A0 is possible

3

[9]

Q20.

(a) (i) $R \sim B(14, 0.35)$

Used somewhere in (a); may be implied

M1

$$P(R \leq 7) = 0.924 \text{ to } 0.925$$

AWFW (0.92466)

A1

2

(ii) $P(R \geq 11) = 1 - P(R \leq 10)$
 $= 1 - (0.9989 \text{ or } 0.9999)$
Requires '1 -' and ≥ 4 dp accuracy

M1

$$= 0.0011$$

AWRT (0.001106)

A1

2

(iii) $P(5 < R < 10) = 0.9940 \text{ or } 0.9989 (p_1)$
Accept 3 dp accuracy
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

minus 0.6405 or 0.4227 (p_2)
Accept 3 dp accuracy

M1

$$= 0.353 \text{ to } 0.354$$

AWFW (0.35346)

A1

or
 B(14, 0.35) expressions stated for **at least 3** terms within $4 \leq R \leq 11$ gives probability
Can be implied by correct answer

(M1)

$$= 0.353 \text{ to } 0.354$$

AWFW (0.35346)

(A2)

(b) $R \sim B(21, 0.35)$

Implied from correct stated formula; do not accept misreads

M1

$$P(R = 4) = \binom{21}{4} (0.35)^4 (0.65)^{17}$$

*Can be implied by a correct answer
Ignore any additional terms*

A1

$$= 0.059 \text{ to } 0.0595$$

AWFW (0.059274)

A1

3

(c) (i) $S \sim B(7, 5/7)$

$$\text{Mean} = np = 7 \times 5/7 = 5$$

If not identified, assume order is μ then σ^2

CAO

B1

$$\text{Variance} = np(1 - p)$$

$$= 7 \times 5/7 \times 2/7 = 10/7 \text{ or } 1.42 \text{ to } 1.43$$

*Must clearly state variance value if
standard deviation (also) stated
CAO / AWFW*

B1

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2

- (ii) **Means** are the **same**
and (both comparisons clearly stated)
Variances/standard deviations are similar
 Do not accept statements involving
 correct/incorrect/exact/etc

*Must have scored B1 B1 in (i) or
B1 B0 plus*

$$10/7 \vee 1.5 \text{ or } \sqrt{10/7} \vee \sqrt{1.5} \text{ **stated**}$$

B1dep

Barry's claim appears/is
sound/valid/correct/likely

Must have scored previous B1dep

B1dep

2

[14]

Q21.

$M \sim B(40, 0.35)$

Used in question; may be implied

M1

(a) $P(M \leq 12) = \underline{\mathbf{0.314}}$

AWRT (0.31431)

A1

2

(b) $P(10 \leq M \leq 15) = \mathbf{0.6946}$ or $\mathbf{0.5721}$ (p_1)

Accept 3 dp rounding or truncation

$$p_2 - p_1 \Rightarrow M0\ M0\ A0$$

$$(1 - p_2) - p_1 \Rightarrow M0\ M0\ A0$$

$$p_1 - (1 - p_2) \Rightarrow M1\ M0\ A0$$

only providing result > 0

M1

minus 0.0644 or 0.1215 (p_2)

Accept 3 dp rounding or truncation

M1

= 0.63(0)

AWRT (0.63021)

A1

OR

$B(40, 0.35)$ expressions stated for **at**

least 3 terms within $9 \leq M \leq 15$ gives probability

Can be implied by correct answer

(M1)

= 0.63(0)

AWRT (0.6302)

(A2)

3

(c) $P(M = 15) = 0.6946 - \mathbf{0.5721}$

Accept 3 dp accuracy

Allow F from (b)

OR

Can be implied by correct answer

M1

$$P(M = 15) = \binom{40}{15} (0.35)^{15} (0.65)^{25}$$

Ignore any additional terms

$$= 0.122 \text{ to } 0.123$$

AWFW (0.12256)

A1

2

[7]

Q22.

Time. $X \sim N(65, 20^2)$

$$(a) \quad (i) \quad P(X < 90) = P\left(Z < \frac{90 - 65}{20}\right) = P\left(Z < \frac{0 - 65}{20}\right) = P(Z < -3.25) = 0.00058$$

Standardising (89.5, 90 or 90.5 or 59.5, 60 or 60.5) with 65 and ($\sqrt{20}$, 20 or 20^2) and/or $(65 - x)$

May be gained in (a)(i) or (a)(ii)

$$= P(Z < 1.25)$$

CAO; ignore inequality and sign

May be implied by a correct answer

M1

A1

$$= 0.893 \text{ to } 0.895$$

AWFW (0.89435)

A1

$$(ii) \quad P(X > 60) = P(Z > -0.25) \\ = P(Z < 0.25)$$

Area change

May be implied by a correct answer or answer > 0.5

M1

$= 0.598 \text{ to } 0.599$
AWFW (0.59871)

A1
5

- (b) (i) $P(1 \text{ in } 6 = 60) = \mathbf{0 \text{ or zero or impossible}}$
Ignore any working
B0 for 'impossible to calculate'

B1
1

- (ii) $P(X < 60) = 1 - [(a)(ii)] \text{ or } (0.401 \text{ to } 0.402)$
May be implied

M1

$P(6 \text{ in } 6 < 60) = p^6 \text{ with } 0 < p < 1$
Any probability to power 6; do not
allow multiplying factors

M1

$= (0.40129)^6 = \mathbf{0.004 \text{ to } 0.00425}$
Dependent on M1 M1 (0.0041759)

A1dep
3

[9]

Q23.

- (a) $M \sim B(50, 0.15)$
Used somewhere in (a); may be implied

M1

- (i) $P(M \leq 10) = \mathbf{0.88(0)}$
AWRT (0.8801)

A1
2

- (ii) $P(M \geq 5) = 1 - P(M \leq 4)$
 $= \mathbf{1 - (0.1121 \text{ or } 0.2194)}$
Requires '1 -'; accept 3 dp accuracy
Implied by 0.888 but not by 0.781

M1

$= \mathbf{0.888}$

AWRT (0.8879)

A1

2

- (iii) $P(6 < R < 12) = \mathbf{0.9372 \text{ or } 0.9699}$ (p_1)
Accept 3 dp accuracy rounding or truncation
 $p_2 - p_1 \Rightarrow M0 M0 A0$
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

minus 0.3613 or 0.2194 (p_2)
Accept 3 dp accuracy

M1

= 0.576
 AWRT (0.5759)

A1

OR
 $B(50, 0.15)$ expressions stated for **at least**
3 terms within $5 \leq R \leq 12$ gives
 probability
Can be implied by correct answer

(M1)

= 0.576
 AFWW (0.5759)

(A2)

3

- (b) $F \sim B(35, 0.11)$
Implied from correct stated formula; do not accept misreads

M1

$$P(F = 4) = \binom{35}{4} (0.11)^4 (0.89)^{31}$$

Can be implied by a correct answer
Ignore any additional terms

A1

= 0.206 to 0.208

AWFW (0.20685)

A1

3

(c) $P(M \text{ and } LH) = 0.52 \times 0.15 = \mathbf{0.078}$

or $N(M) = 2000 \times 0.52 = \mathbf{1040}$

≥ 1 of these 2 probabilities or ≥ 1 of these 2 numbers attempted; may be implied

M1

$P(F \text{ and } LH) = 0.48 \times 0.1 = \mathbf{0.0528}$

or $N(F) = 2000 \times 0.48 = \mathbf{960}$

2 probabilities or 2 numbers evaluated correctly

A1

$N(M \text{ and } LH) = 2000 \times 0.078 = 1040 \times 0.15 = \mathbf{156}$

$N(F \text{ and } LH) = 2000 \times 0.0528 = 960 \times 0.11 = \mathbf{105.6}$

Evaluation of ≥ 1 of these 2 numbers

or

$P(LH) = 0.078 + 0.0528 = \mathbf{0.1308}$

or

Addition of these 2 probabilities

A1

$N(LH) = 156 + 105.6 = 2000 \times 0.1308$

= 261 to 262

262/2000 $\Rightarrow$ A0

AWFW (261.6)

A1

4

[14]

Q24.

(a) B (50, 0.15)

Used at least once in (a)

M1

(i) $P(E = 6) = 0.3613 - 0.2194$

Accept 3 dp accuracy

or

$$= \binom{50}{6} (0.15)^6 (0.85)^{44}$$

Either expression

M1

$$= 0.142$$

AWRT (0.1419)

A1

3

(ii) $P(6 < E \leq 12) = 0.9699 \text{ or } 0.9372$

Accept 3 dp accuracy / truncation

M1

minus 0.3613 or 0.2194

Accept 3 dp accuracy

M1

$$= 0.608 \text{ to } 0.609$$

AWFW (0.6086)

A1

OR

B(50, 0.15) expressions stated for at

least 3 terms within $6 \leq E \leq 12$ gives probability

Or implied by a correct answer

(M1)

$$= 0.608 \text{ to } 0.609$$

AWFW

(A2)

3

(b) Mean, $\mu = np = 80 \times 0.15 = 12$

CAO

B1

Variance, $\sigma^2 = np(1 - p)$

$$= 80 \times 0.15 \times 0.85 = 10.2$$

CAO; or equivalent

B1

2

[8]

Q25.

- (a) (i) B(16 or 25 or 40, 0.45)
Used at least once in (a)(i) to (iii)

M1

$$P(S = 3) = \binom{16}{3} (0.45)^3 (0.55)^{13}$$

*May be implied by correct answer
 Ignore any additional terms*

A1

$$= 0.021 \text{ to } 0.022$$

AWFW (0.0215)

A1

3

- (ii) $P(S < 10) = 0.3843 \text{ or } 0.2424$
Accept 3 dp accuracy from tables or calculation

B1

$$= 0.242 \text{ to } 0.243$$

AWFW (0.2424)

B1

2

- (iii) $P(15 \leq S \leq 20) = 0.7870 \text{ or } 0.6844 \quad (p_1)$
Accept 3 dp accuracy

M1

$$p_2 - p_1 \Rightarrow M0 \ M0 \ A0$$

$$p_1 - (1 - p_2) \Rightarrow M1 \ M0 \ A0$$

$$\text{minus } 0.1326 \text{ or } 0.2142 \quad (p_2)$$

Accept 3 dp accuracy / truncation

M1

$$= 0.654 \text{ to } 0.655$$

AWFW (0.6544)

A1

OR

B(40, 0.45) expressions stated for at least 3 terms within $14 \leq S \leq 20$ gives probability

Or implied by a correct answer

(M1)

$$= 0.654 \text{ to } 0.655$$

AWFW

(A2)

3

(iv) Mean, $\mu = np = 50 \times 0.45 = 22.5$ or $22\frac{1}{2}$

CAO (22.5 = 22 or 23 $\Rightarrow$ ISW)

B1

Variance, $\sigma^2 = np(1 - p)$

Accept $12\frac{3}{8}$ or $\frac{99}{8}$

$$= 50 \times 0.45 \times 0.55$$

$$= 12.3 \text{ to } 12.4$$

AWFW (12.375)

B1

2

- (b) (i) **Non-independence** of senior citizens travel
 Senior citizens tend to **travel in pairs/groups**

*Or equivalent; but must be a clear
 indication of non-independent events*

B1

1

- (ii) 7.15 am is outside 9.30 am to 11.30 am
 Cannot use SCPs before 9.30 am
 Cannot use SCPs @ 7.15 am
 Cannot use SCPs during morning 'rush hour'
 Value of p likely to be smaller/different/zero
 Data not available

Or equivalent

*Accept other **sensible** reasons*

B1

Senior citizens not out at this time

Passengers likely to be workers/school children

*Distribution of **types of** passenger different*

1

[12]

Q26.

(a) $R \sim B(50, 0.15)$

(i) $P(R < 10) = \mathbf{0.791}$
AWRT (0.7911)

B1

(ii) $P(5 \leq R \leq 10) = 0.8801$ or 0.7911 (p_1)
Accept 3 dp accuracy
 $(1 - p_2) - p_1 \Rightarrow M0 M0 A0$
 $p_1 - (1 - p_2) \Rightarrow M1 M0 A0$
only providing result > 0

M1

minus 0.1121 or 0.2194 (p_2)
Accept 3 dp accuracy

M1

$= 0.768$
AWRT (0.7680)

A1

or
 $B(50, 0.15)$ expressions stated for **at least 3**
 terms within $4 \leq R \leq 10$ gives probability
Can be implied by correct answer

(M1)

$= 0.768$
AWRT

(A2)

4

(b) **Confusion of 22, 35, 120 and/or 0.15, 0.06**
*Do **not** treat as misreads*

(i) $S \sim B(22, 0.06)$
*Used in (b)(i) as evidenced by any
 correct binominal term for $S > 0$*

M1

$P(S = 2) = \binom{22}{2} (0.06)^2 (0.94)^{20}$
Can be implied by correct answer

Ignore any additional terms

A1

= 0.24 to 0.242
AWFW (0.24125)

A1

3

- (ii) $P(S \geq 1) = 1 - q^{35}$ where $0.84 \leq q \leq 0.96$
Can be implied by correct answer
Award for $(0.94)^{35}$ seen in an
expression but not if accompanied
by a multiplier $\neq 1$

M1(B1)

= 0.885 to 0.89
AWFW (0.88532)

A1

2

- (iii) Mean = $np = 120 \times 0.94 = 112.8$ or 113
Either

B1

If not identified, assume order is μ then σ^2
 Variance = $np(1 - p)$
Must clearly state variance value

= $120 \times 0.94 \times 0.06 = 6.76$ to 6.78
AWFW (6.768)

B1

2

- (iv) **Means** are (approximately) the **same** stated
Must have scored 1st B1 in (iii)

or
Variances are (very) **different** stated
Must have scored 2nd B1 in (iii)

B1

Agree with $P(\text{sorts letter incorrectly}) = 0.06$
Dependent on 'means same' stated

B1dep

Disagree with independent from letter to letter
Dependent on 'variances different' stated

B1dep

3

[14]

Q27.

- (a) Use of binomial in (a) or (b)
Can be implied by answers

M1

(i) $P(X \geq x) = 1 - P(X \leq x - 1)$

OR

$= 1 - B(\Sigma x, 50, 0.08)$

Identified from an answer / 1 - answer
Can be implied from a correct answer
Identified from an answer/expression

M1

$= 1 - 0.0827 = 0.915 \text{ to } 0.92(0)$
AWFW (0.9173)

A1

$\geq 1 \text{ correct} \Rightarrow M1 M1$

(ii) $P(X \geq 3)$
 $= 1 - 0.2260 = 0.77(0) \text{ to } 0.775$
AWFW (0.7740)

A1

4

- (b) (i) $P(Y = 0) = (1 - 0.025)^{15} = 0.975^{15}$
Can be implied from correct answer

M1

$= 0.68(0) \text{ to } 0.685$
AWFW (0.6840)

A1

- (ii) $P(Y \geq 1) = 1 - (i)$
Can be implied from answer if $\epsilon (0, 1)$

M1

$$= 0.315 \text{ to } 0.32(0)$$

ft on (i) if $\varepsilon (0, 1)$ (0.3160)

A1ft

4

(c) Probability = [(b)(ii) $\times$ (a)(i)] or (0.316×0.917)

M1

$$[(b)(i) \times (a)(ii)] \text{ or } (0.684 \times 0.774)$$

M1

Ignore additional terms

$$= 0.2898 + 0.529$$

2 terms added with ≥ 1 correct

A1

$$= 0.81 \text{ to } 0.83$$

AWFW (0.8193)

A1

4

[12]

Q28.

Binomial distribution

Used somewhere in question

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M1

(a) $M \sim B(40, 0.35)$

Used; may be implied

A1

$$P(M \leq 15) = 0.69(0) \text{ to } 0.696$$

AWFW (0.6946)

A1

3

(b) $W \sim B(10, 0.29)$

Used; may be implied

B1

$$P(W = 3) = \binom{10}{3} (0.29)^3 (0.71)^7$$

Stated; may be implied

M1

$$= 0.266 \text{ to } 0.2665$$

AWFW (0.2662)

Note: $B(10, 0.3) \Rightarrow 0.2668$

A1

3

(c) (i) $n = 20 \quad p = 0.71$

Stated or used; may be implied by 14.2

B1

Mean, $\mu = np = 14.2$

CAO

B1

Variance, $\sigma^2 = np(1 - p)$
 $= 4.11 \text{ to } 4.12$

AWFW (4.118)

B1

3

- (ii) Mean of 16.5 is greater/different **or**
 $16.5/20 = 0.825$ is greater/different to 0.71

Dependent on $\mu = 14.2$

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Means and variances are different

(B2, 1 dep)

Variance of 2.50 is smaller/different

Dependent on $\sigma^2 = 4.11 \text{ to } 4.12$

B1dep

Suggests **claim** that groups are not
 random samples **is justified**

*Dependent on previous 2 marks
 Or equivalent*

B1dep

3

Q29.

Binomial distribution

Used somewhere in question

M1

(a) (i) $M \sim B(40, 0.35)$

Used; may be implied

A1

$$P(M \leq 15) = 0.69(0) \text{ to } 0.696$$

AWFW (0.6946)

A1

3

(ii) $P(10 < M < 20) = 0.9637 \text{ or } 0.9827$

Accept 3 dp accuracy

M1

minus 0.1215 or 0.0644

Accept 3 dp accuracy

M1

$$= 0.84(0) \text{ to } 0.843$$

AWFW (0.8422)

A1

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OR

$B(40, 0.35)$ expressions stated for

at least 3 terms within $10 \leq M \leq 20$

Answer = 0.84(0) to 0.843

Or implied by a correct answer

(M1)

AWFW

(A2)

3

(b) $W \sim B(10, 0.29)$

Used; may be implied

B1

$$P(W = 3) = \binom{10}{3} (0.29)^3 (0.71)^7$$

Stated; may be implied

M1

$$= 0.266 \text{ to } 0.2665$$

AWFW (0.2662)

Note: $B(10, 0.3) \Rightarrow 0.2668$

A1

3

(c) (i) $n = 20$ $p = 0.71$

Stated or used; may be implied by 14.2

B1

$$\text{Mean, } \mu = np = 14.2$$

CAO

B1

$$\text{Variance, } \sigma^2 = np(1 - p) = 4.11 \text{ to } 4.12$$

AWFW (4.118)

B1

3

- (ii) Mean of 16.5 is greater/different **or**
 $16.5/20 = 0.825$ is greater/different to 0.71

Dependent on $\mu = 14.2$

B1dep

Means and variances are different

(B2, 1 dep)

Variance of 2.50 is smaller/different

Dependent on $\sigma^2 = 4.11 \text{ to } 4.12$

B1dep

Suggests **claim** that groups are not
 random samples **is justified**

*Dependent on previous 2 marks
 Or equivalent*

B1dep

Q30.

(a)

		—————	E(0.30)	0.0750
	—————	B(0.25)	—————	M(0.50) 0.1250
	—————	W(0.20)		0.0500
		—————	E(0.40)	0.0240
	—————	S(0.60)	—————	M(0.45) 0.2700
	—————	W(0.15)		0.0900
		—————	E(0.55)	0.0825
	—————	D(0.15)	—————	M(0.35) 0.0525
	—————	W(0.10)		0.0150

B, S & D with 3 probabilities

B1

3 × (E, M & W) each with 3 probabilities

B2

≥ 1 × (E, M & W) (each) with 3 probabilities

(B1)

3

(b) (i) $P(E) = (0.25 \times 0.3) + (0.6 \times 0.4) + (0.15 \times 0.55)$
 $= 0.075 + 0.24 + 0.0825 =$
 $\geq 1 \text{ term correct}$

M1

0.397 to 0.398 or 159/400
AWFW/CAO (0.3975)

A1

2

(ii) $P(D | E) = \frac{0.0825}{(b)(i)} =$
Or equivalent

M1

0.207 to 0.208 or 11/53
AWFW/CAO (0.2075)

A1

(c) $X \sim B(10, (b) (ii))$

Used

M1

$$P(X = 4) = \binom{10}{4} (0.2075)^4 (0.7925)^6 =$$

ft on (b)(ii)

A1ft

0.0955 to 0.0975

AWFW (0.09645)

A1

3

[10]

Q31.

(a) Use of binomial in (a), (b) or (c)

Can be implied

M1

$$P(P = 5) = \binom{12}{5} p^5 (1-p)^7$$

Allow $p = 0.45, 0.30, 0.22$ or $\frac{1}{3}$

M1

= 0.207 to 0.208

AWFW (0.2075)

A1

3

(b) (i) $P(10 < S < 15) = 0.8074$ or 0.8849

Allow 3dp accuracy

M1

minus 0.3087 or 0.4406

Allow 3dp accuracy

M1

= 0.498 to 0.499
AWFW (0.4987)

A1

or
 $B(40 \ 0.30)$ expressions stated for
at least 3 terms within $10 \leq B' \leq 20$
Or implied by a correct answer

(M1)

Answer = 0.498 to 0.499
AWFW

(A2)

3

- (c) Mean, $\mu = np = 22$
CAO; ft on p only ($0 < p < 1$)

B1ft

Variance, $\sigma^2 = np(1 - p) = 17.16$
Use of $np(1 - p)$ even if SD

M1

Standard deviation = $\sqrt{17.1 \text{ to } 17.2}$
 or = 4.14 to 4.15
AWFW

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A1

3

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[9]

Q32.

- (a) Use of binomial in (a), (b) or (c)
Can be implied

M1

$$P(X = 5) = \binom{16}{5} p^5 (1 - p)^{11}$$

Allow $p = 0.45, 0.25, 0.30$ or $\frac{1}{3}$

M1

$$= 0.112$$

AWRT (0.1123)

A1

3

(b) (i) $B(50, 0.25)$

Used; can be implied

B1

$$P(C \leq 12) = 0.511$$

AWRT (0.5110)

B1

2

(ii) $P(10 < B' < 20) = 0.9152 \text{ or } 0.9522$

Allow 3 dp accuracy

M1

minus 0.0789 or 0.1390

Allow 3 dp accuracy

M1

$$= 0.836$$

AWRT (0.8363)

A1

or

$B(50, 0.30)$ expressions stated for **at least 3** terms
within $10 \leq B' \leq 20$

Or implied by a correct answer

(M1)

$$\text{Answer} = 0.836$$

AWRT

(A2)

3

(c) $n = 40, p = 0.7$

Both used; can be implied

B1

$$\text{Mean } \mu = np = 28$$

CAO; ft on p only

B1ft

Variance $\sigma^2 = np(1 - p) = 8.4$
Use of $np(1 - p)$ even if SD

M1

Standard deviation = $\sqrt{8.4}$
 or = 2.89 to 2.9
 CAO; AFWW

A1

4

[12]

Q33.

(a) Use of binomial in (a) or (b)
PI

M1

$$P(R_7 = 3) = \binom{7}{3} (0.45)^3 (0.55)^4$$

Correct expression

A1

or
 = 0.6083 – 0.3164
 = 0.29 to 0.292
 AFWW (0.2919)

A1

3

(b) (i) $P(R_{30} < 15) = 0.64$ to 0.645
 AFWW (0.6448)

B2

SC: Answer = 0.769 to 0.77
 AFWW (0.7691)

(B1)

2

(ii) $P(R_{30} > 10) = 1 - 0.135$
PI

M1

= 0.86 to 0.87

AWFW (0.8650)

A1

SC: Answer = 0.93 to 0.931

AWFW (0.9306)

(B1)

2

(iii) $P(12 \leq R_{30} \leq 18) = 0.9666 \text{ or } 0.9286$

Allow 3 dp accuracy

M1

minus 0.2327 or 0.3592

Allow 3 dp accuracy

M1

= 0.73 to 0.734

AWFW (0.7339)

A1

OR

B(30, 0.30) expressions stated for at least
3 terms within $12 \leq R_{30} \leq 18$

Or implied by a correct answer

(M1)

Answer = 0.73 to 0.734

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(A2)

3

[10]

Q34.

(a) Use of binomial in (a) or (b)(i)

PI

M1

(i) $P(T_{10} \leq 3) = 0.38 \text{ to } 0.383$

AWFW (0.3823)

B1

2

- (iii) $P(10 < T_{40} < 20) = 0.8702$ or 0.9256
Allow 3 dp accuracy

M1

minus 0.0352 or 0.0156
Allow 3 dp accuracy

M1

$= 0.83$ to 0.84
AWFW (0.835)

A1

OR

B $(40, 0.40)$ expressions stated for at least
 3 terms within $10 \leq T_{40} \leq 20$
Or implied by a correct answer

(M1)

Answer $= 0.83$ to 0.84
AWFW

(A2)

3

- (b) (i) $n = 5$ $p = 0.4$

Mean, $\mu = np = 2$
 CAO

B1

Variance, $\sigma^2 = np(1 - p) = 1.2$
Use of $np(1 - p)$ even if SD

M1

Standard deviation $= \sqrt{1.2}$
 or $= 1.09$ to 1.1
 CAO
 AFWW

A1

3

- (ii) Mean $(\bar{x}) = 2$

$$CAO \sum x = 26$$

B1

Standard Deviation (s_n, s_{n-1})
= 1.1 to 1.16

$$\sum x^2 = 68$$

AWFW (1.1094 or 1.1547)

B2

If neither correct but use of mean $(\bar{x}) = \frac{\sum x}{13}$

(M1)

3

- (iii) Means are same and SDs are similar/same
Means are same but SDs are different

Must have scored full marks in (b) (i) and (b) (ii)

B1

so

↑Dep↑

Trina's claims appear valid / invalid

B1

2

[13]

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Q35.

- (a) (i) $B(20, p)$

Use of in (a)

M1

$$P(O < 10 \mid p = 0.40) = 0.872 \text{ to } 0.873$$

AWFW 0.8725

A1

2

$$(ii) P(A = 8) = \binom{20}{3} (p)^3 (1 - p)^{17}$$

Any value of p with $0 < p < 1$

M1

$$P(A = 8 \mid p = 0.20) =$$

$$\binom{20}{3} (0.28)^3 (0.72)^{17}$$

Fully correct expression

A1

$$= 0.093 \text{ to } 0.095$$

AWFW 0.0940

A1

3

(iii) $P(4 < B < 8 \mid p = 0.20) =$

$$P(B \leq 7 \text{ or } 8) = 0.9679 \text{ or } 0.9900$$

M1

$$\text{minus } P(B \leq 4 \text{ or } 3) = 0.6296 \text{ or } 0.4114$$

M1

$$= 0.338 \text{ to } 0.34$$

AWFW 0.3383

A1

or

B(20, 0.2) expressions stated for at least 3 of
B = 4, 5, 6, 7 & 8

(M1)

Answer

Or implied by a correct answer

(A2)

3

(b) Mean, $\mu = np = 500 \times 0.12 = 60$

CAO

B1

$$\text{Variance, } \sigma^2 = np(1 - p) = 60 \times 0.88 = 52.8$$

CAO

B1

2

[10]